

1. What did you accomplish this week?

I done the basic funding analysis for Northeastern's grant data (2,670 unique grants and \$2.15 billion in total funding). I ran five analyses: funding concentration, college-level breakdowns (raw and tenure track faculty normalized), agency analysis, trends over time, co-PI structure, and a gap analysis on tenure-track faculty.

2. Did you learn or discover anything new?

Funding is more concentrated than expected. The Gini coefficient came out at 0.65, meaning the distribution is very unequal. Just 173 out of 532 PIs (32%) account for 80% of all funding. The top PI alone, Akram Alshawabkeh in Engineering, has brought in \$83.6M over his career.

Bouvé College of Health Sciences punches well above its weight. In raw totals it ranks third (\$367M), but once you normalize by tenure-track faculty it has the second-highest funding per person (\$4.2M per tenure-track faculty), ahead of Engineering. This is almost entirely driven by NIH, Bouvé's co-PI rate is the lowest of any college (12.5%), meaning most of its faculty are lead PIs on large independent grants rather than joining other people's projects.

Collaborative grants are meaningfully larger. Grants with at least one co-PI average \$1.09M vs. \$722K for solo PI grants, about 51% larger.

The gap between funded and unfunded tenure-track faculty is larger than expected in some colleges. D'Amore-McKim Business has 93 tenure-track faculty but only 7 (7.5%) appear in the grant data. Arts, Media and Design and Law are similar, 83.5% and 87.9% of their tenure-track faculty have no grant record. By contrast, Khoury (80%) and Engineering (79%) have the majority of their tenure-track faculty funded.

NSF dominates in grant count (63%) but NIH dominates in average grant size (\$1.53M per grant vs. \$624K for NSF). So the two agencies are almost inverses of each other, NSF funds many smaller grants broadly across campus, NIH funds fewer but much larger grants concentrated in Bouvé and Science.

3. Am I uncertain or stuck about anything?

Some of the "unfunded" faculty may simply be newer hires who haven't had time to secure a grant yet. The data doesn't let us separate those cases from faculty who have been here for years without external funding.

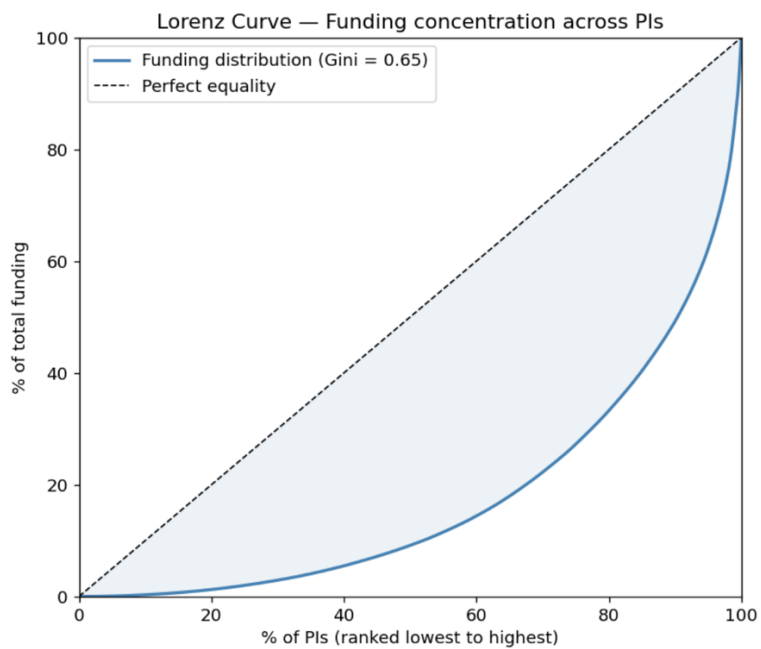
Also, College of Professional Studies shows 100% funded (1 of 1 tenure-track faculty), that's a sample size of one and shouldn't be interpreted as a real signal.

4. Plan for next week

Next week I'll focus more on topic modeling for the grant abstracts. The goal is to identify the main research themes across funded grants. I'll run LDA, NMF, and BERTopic, compare the results, and label the topics.

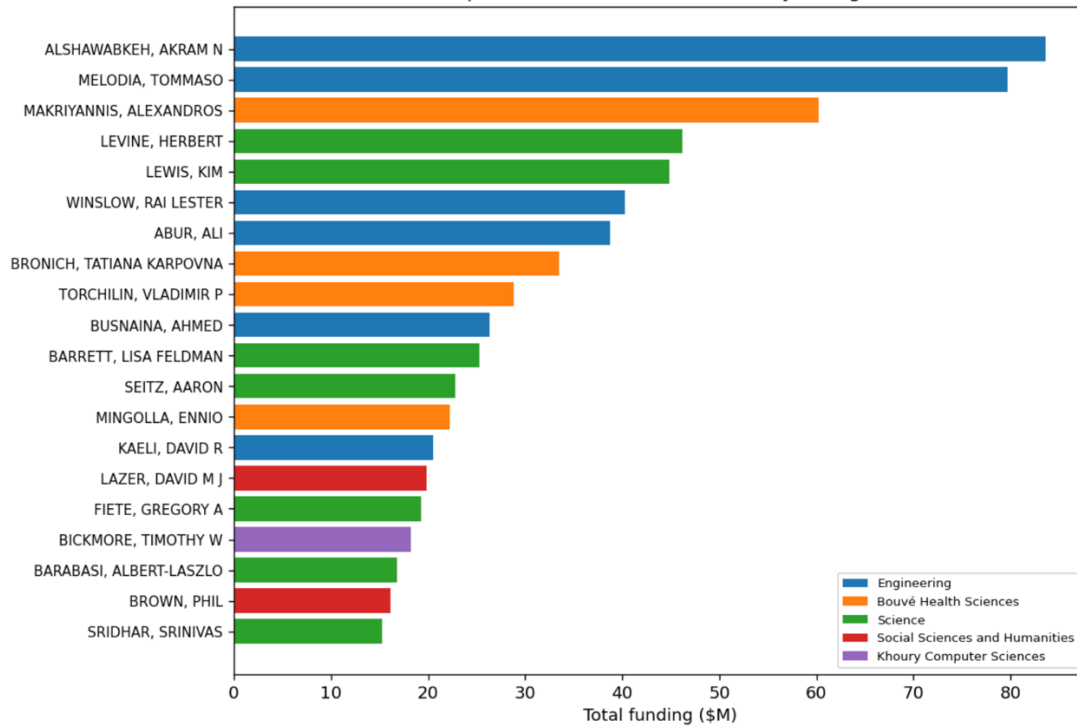
Figures and summary:

Total unique grants: 2,670
Total funding (deduped): \$2,149.0M
Unique PIs: 532
PIs for top 80% of funding: 173 out of 532
Top agency (NSF): 63.0% of grants
Top college (Engineering): 771.9M total
Collaborative grants: 598 (22.4%)
Tenure-track with grants: 486 out of 958 (50.7%)

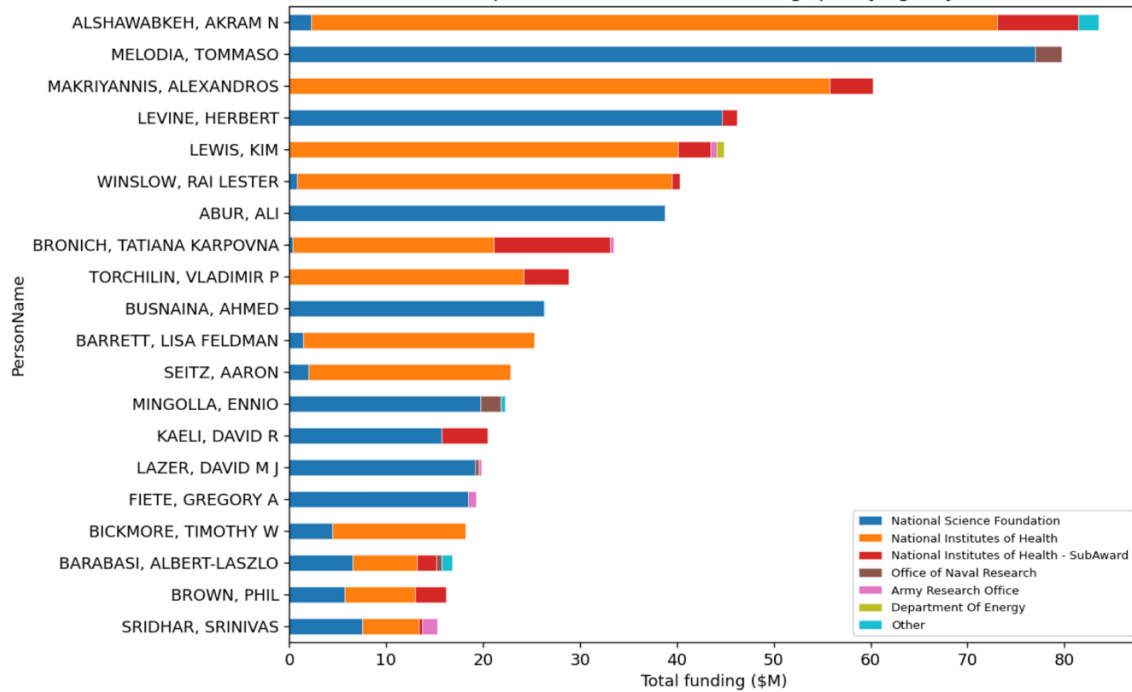


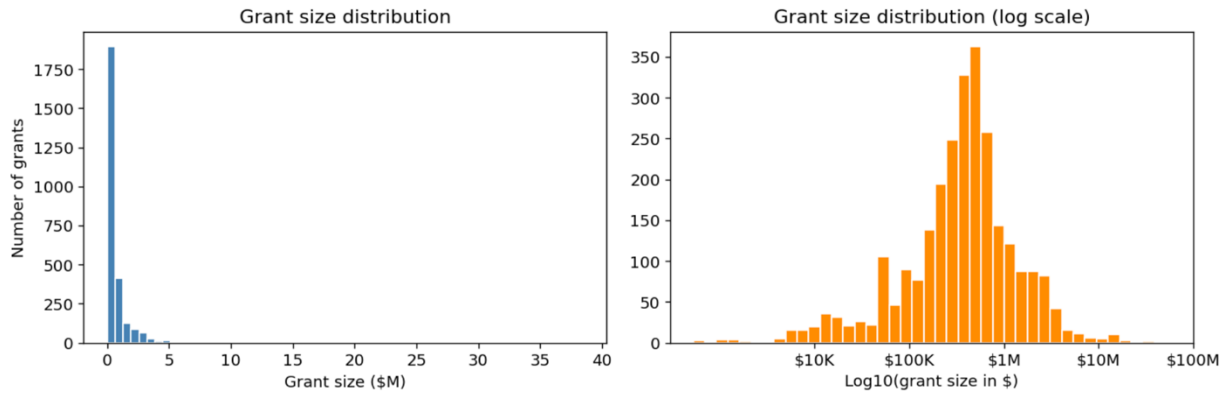
Gini coefficient: 0.646 (0 = perfect equality, 1 = one PI has everything)

Top 20 tenure-track PIs colored by college



Top 20 tenure-track PIs — funding split by agency





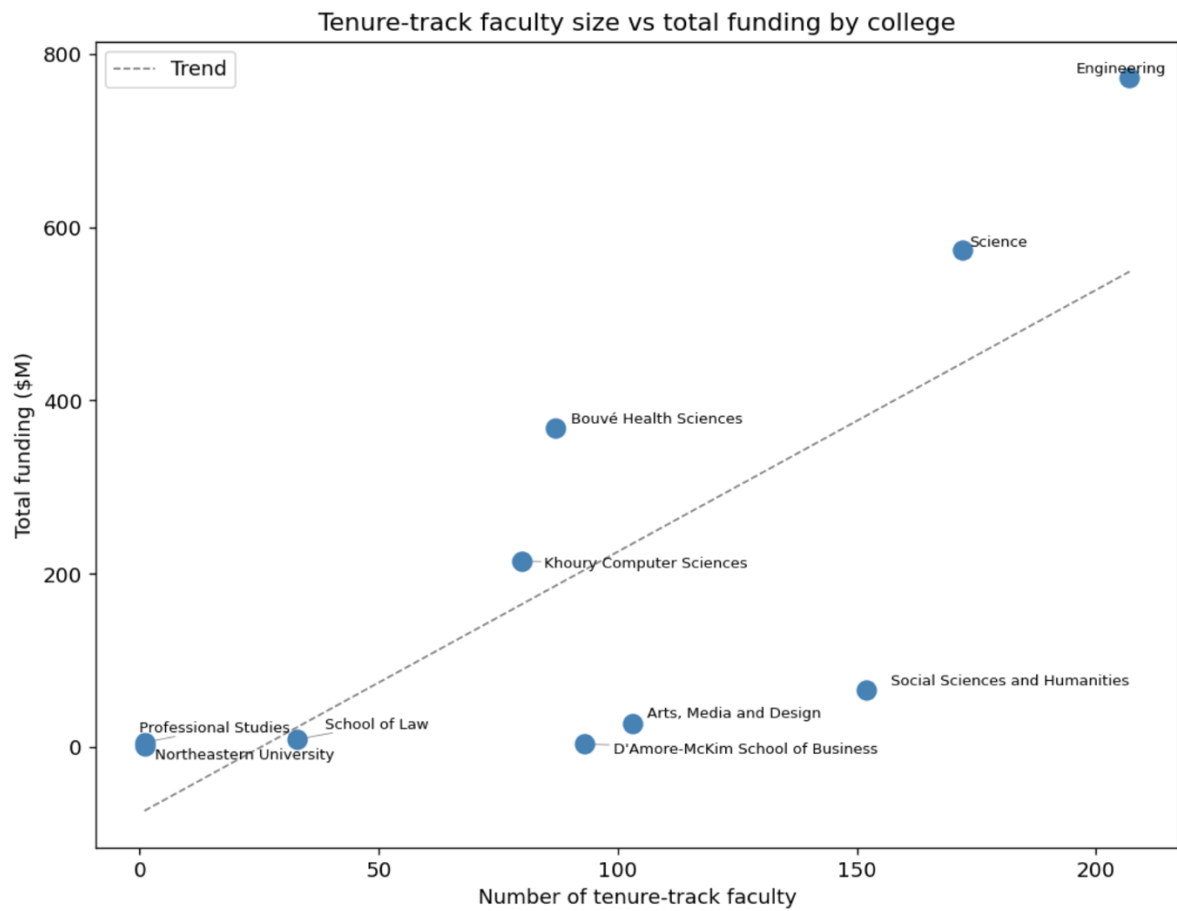
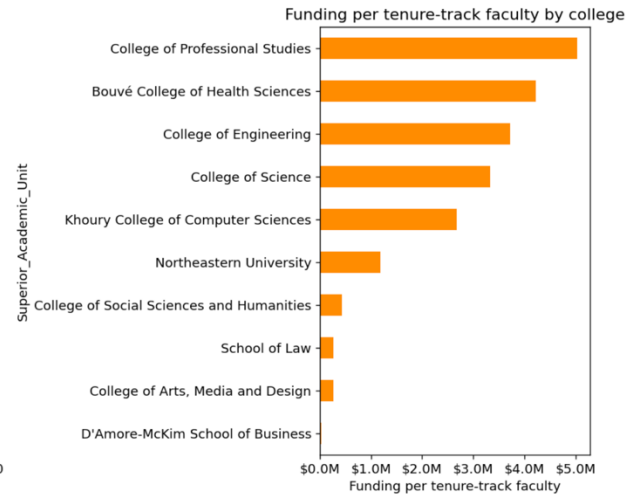
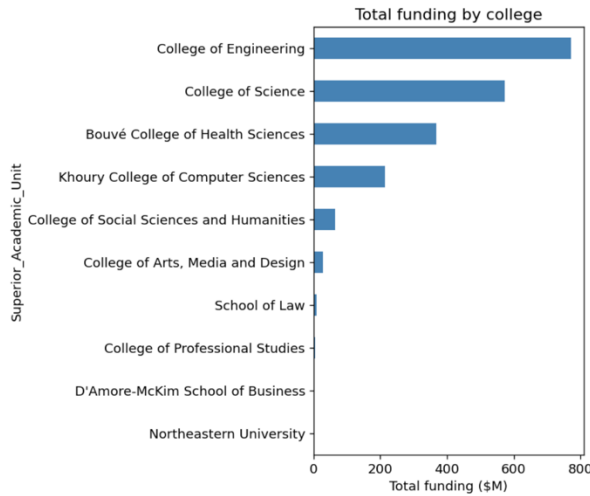
Grant size percentiles:
 10th percentile: \$50,000
 25th percentile: \$190,386
 50th percentile: \$404,666
 75th percentile: \$739,390
 90th percentile: \$1,758,103
 95th percentile: \$2,690,955
 99th percentile: \$7,283,008

Tenure-track faculty per college:

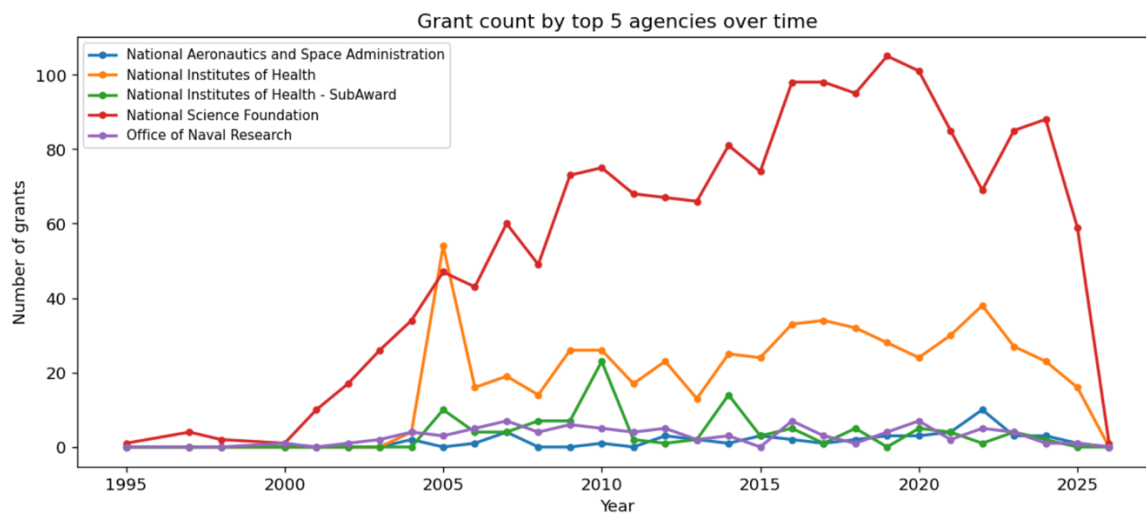
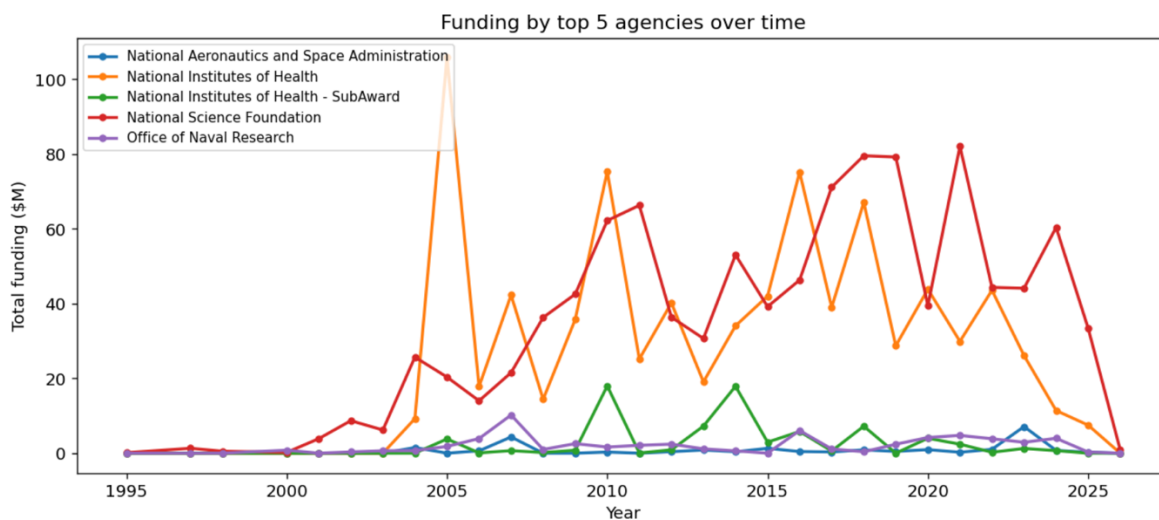
College of Engineering	207
College of Science	172
College of Social Sciences and Humanities	152
College of Arts, Media and Design	103
D'Amore-McKim School of Business	93
Bouvé College of Health Sciences	87
Khoury College of Computer Sciences	80
School of Law	33
Mills College at Northeastern	29
College of Professional Studies	1
Northeastern University	1

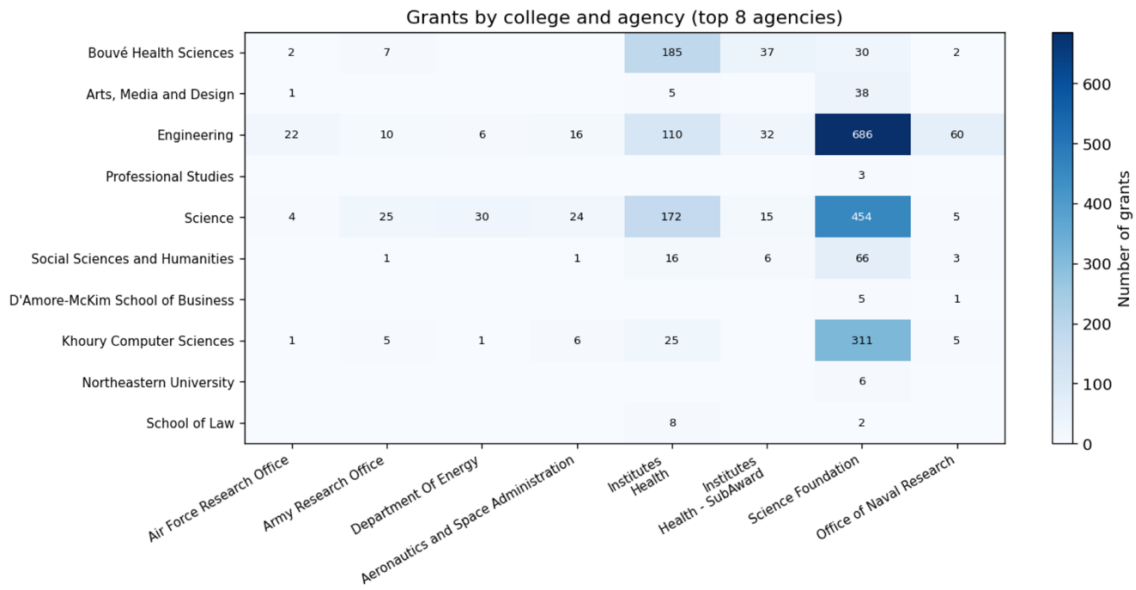
Normalized metrics per college:

	total_faculty	tenure_faculty	num_grants	total_M	dollars_per_faculty	dollars_per_tenure	grants_per_tenure
Superior_Academic_Unit							
College of Professional Studies	110	1	4	5.04	45,796.84	5,037,652.00	4.00
Bouvé College of Health Sciences	269	87	275	367.71	1,366,941.42	4,226,520.02	3.16
College of Engineering	356	207	957	771.89	2,168,230.50	3,728,937.48	4.62
College of Science	332	172	749	573.73	1,728,096.86	3,335,628.83	4.35
Khoury College of Computer Sciences	210	80	359	214.49	1,021,371.62	2,681,100.51	4.49
Northeastern University	2	1	6	1.19	595,022.50	1,190,045.00	6.00
College of Social Sciences and Humanities	304	152	115	65.37	215,042.42	430,084.85	0.76
School of Law	58	33	10	8.98	154,890.00	272,230.91	0.30
College of Arts, Media and Design	210	103	46	27.65	131,679.09	268,471.92	0.45
D'Amore-McKim School of Business	215	93	9	3.35	15,560.64	35,973.53	0.10



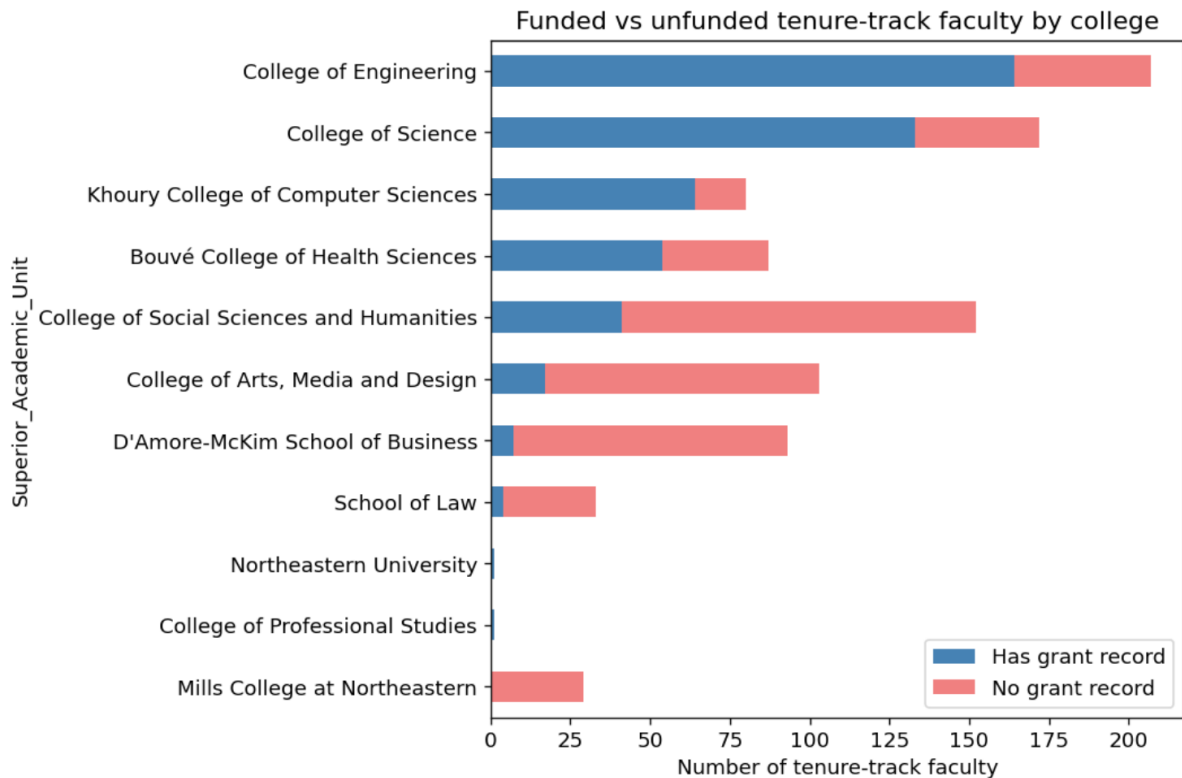
AgencyName	num_grants	pct_grants	total_M	pct_dollars	avg_grant
National Science Foundation	1682	63.00	1,049.40	48.80	623,901.65
National Institutes of Health	546	20.40	833.00	38.80	1,525,633.78
National Institutes of Health - SubAward	104	3.90	74.60	3.50	717,271.26
Office of Naval Research	87	3.30	59.69	2.80	686,084.85
National Aeronautics and Space Administration	49	1.80	22.06	1.00	450,124.69
Army Research Office	48	1.80	13.18	0.60	274,675.54
Department Of Energy	37	1.40	33.84	1.60	914,595.76
Air Force Research Office	31	1.20	11.45	0.50	369,284.68
Dept. Health and Human Services	21	0.80	32.48	1.50	1,546,518.05
National Endowment for the Humanities	19	0.70	2.01	0.10	106,041.58
National Oceanic and Atmospheric Administration	17	0.60	5.52	0.30	324,958.24
Department Of Education	8	0.30	4.39	0.20	548,708.38
Department of Agriculture	8	0.30	2.14	0.10	267,578.12
Environmental Protection Agency	3	0.10	3.32	0.20	1,108,213.33
The Wellcome Trust	3	0.10	0.51	0.00	171,529.00
Biotechnology and Biological Sciences Research Council	2	0.10	0.70	0.00	351,947.00
Medical Research Council	1	0.00	0.34	0.00	338,762.00
Institute of Museum and Library Services	1	0.00	0.03	0.00	25,000.00
Economic and Social Research Council	1	0.00	0.03	0.00	31,615.00
Department of Veterans Affairs	1	0.00	0.15	0.00	153,209.00
Arts and Humanities Research Council	1	0.00	0.15	0.00	148,451.00





Funded vs unfunded tenure-track faculty per college:

	funded	total	unfunded	funded_pct	unfunded_pct
Superior_Academic_Unit					
College of Professional Studies	1	1	0	100.00	0.00
Northeastern University	1	1	0	100.00	0.00
Khoury College of Computer Sciences	64	80	16	80.00	20.00
College of Engineering	164	207	43	79.20	20.80
College of Science	133	172	39	77.30	22.70
Bouvé College of Health Sciences	54	87	33	62.10	37.90
College of Social Sciences and Humanities	41	152	111	27.00	73.00
College of Arts, Media and Design	17	103	86	16.50	83.50
School of Law	4	33	29	12.10	87.90
D'Amore-McKim School of Business	7	93	86	7.50	92.50
Mills College at Northeastern	0	29	29	0.00	100.00



Grants per funded tenure-track PI:

```
count    532.00
mean       5.02
std        4.72
min         1.00
25%         2.00
50%         3.00
75%         7.00
max        35.00
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Name: num_grants, dtype: float64

Distribution:

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>= 1 grants : 532 people
>= 2 grants : 412 people
>= 5 grants : 215 people
>= 10 grants : 78 people
>= 20 grants : 9 people
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